

Spectral Fixed Point Theory 2.0: Core Derivation via a Zero-Free Wall

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December 30, 2025

1 Preliminaries

Let $\zeta(s)$ be the Riemann zeta function and let $\rho = \beta + i\gamma$ denote its non-trivial zeros (counted with multiplicity). Define

$$\beta_0 := \sup_{\rho} \operatorname{Re}(\rho) \in \left[\frac{1}{2}, 1\right].$$

The Riemann Hypothesis is equivalent to $\beta_0 = \frac{1}{2}$.

For $b \in [\frac{1}{2}, 1)$, we take $\arg \zeta(b+it)$ to mean the imaginary part of a branch of $\log \zeta$ defined along the vertical line $\Re(s) = b$ by continuous variation (with the standard convention that zeros/pole crossings are handled via the contour method used in [SBM09]).

2 The SBM Master Identity and Instability Mass

2.1 Contour integral setup

Sekatskii, Beltraminelli, and Merlini (SBM) [SBM09] consider a rectangular contour C and the contour integral

$$I = \oint_C g(z) \log \zeta(z) dz, \quad g(z) := -i \frac{z - b}{(c^2 - (z - b)^2)(d^2 - (z - b)^2)},$$

where $c, d > 0$ and $c \neq d$. Evaluating I by residues and comparing with its evaluation on the left edge $z = b + it$ yields an identity involving an integral of $\arg \zeta(b + it)$ and an explicit sum over zeros to the right of the wall $\Re(s) = b$.

2.2 Identity with explicit zero term

Define the weight function

$$W_{c,d}(t) := \frac{t}{(c^2 + t^2)(d^2 + t^2)}.$$

SBM obtain an equality of the form (their Eq. (1), specialized to $b < 1$)

$$\int_0^\infty W_{c,d}(t) \arg \zeta(b + it) dt = \mathcal{C}_{\text{expl}}(b; c, d) - S(b; c, d), \quad (1)$$

where $\mathcal{C}_{\text{expl}}(b; c, d)$ is the explicit closed-form expression given below, and

$$S(b; c, d) = \sum_{\rho: \text{Re}(\rho) > b} \left(\text{explicit contribution from } \rho \right),$$

is a sum over *exactly those zeros with* $\text{Re}(\rho) > b$ (multiplicity included, conjugates paired). SBM emphasize that the sum is taken over zeros $\rho_k = \sigma_k + it_k$ with $b < \sigma_k$ and that conjugate pairs are combined to yield a real contribution. [SBM09, see Eq. (1) and surrounding discussion]

2.3 The explicit closed-form constant

The constant $\mathcal{C}_{\text{expl}}(b; c, d)$ is given by

$$\mathcal{C}_{\text{expl}}(b; c, d) := \frac{\pi}{2(d^2 - c^2)} \ln \left| \frac{\zeta(b + d)}{\zeta(b + c)} \right| + \frac{\pi}{2(d^2 - c^2)} \ln \left| \frac{(d^2 - (1 - b)^2) c^2}{(c^2 - (1 - b)^2) d^2} \right|. \quad (2)$$

This expression arises from the residues at the poles of the kernel $g(z)$ at $z = b + c$, $z = b + d$, and the contribution from the pole of ζ at $z = 1$, as derived in [SBM09].

[On positivity/sign control] SBM discuss sufficient conditions under which each off-wall zero contributes with the same sign, by comparing c, d to the smallest ordinate $t_{\min}$ of any hypothetical zero with $\Re(\rho) > 1/2$. They then give an explicit numerical admissibility regime for (c, d) based on the verified height range available at the time of writing [SBM09]. This sign discussion is logically separate from the core equivalence (the integral equality holds iff there are no zeros with $\Re(\rho) > b$), but it motivates interpreting $S(b; c, d)$ as an “instability mass.”

Definition 2.1 (Volchkov–Sekatskii divergence). Fix admissible parameters (c, d) . Define the divergence as the absolute deviation from the SBM equality:

$$\Delta_V(b; c, d) := \left| \mathcal{C}_{\text{expl}}(b; c, d) - \int_0^\infty W_{c,d}(t) \arg \zeta(b + it) dt \right|.$$

By (1), one has $\Delta_V(b; c, d) = |S(b; c, d)|$, and in the same-sign regime discussed by SBM one further has $\Delta_V(b; c, d) = S(b; c, d) \geq 0$.

3 The theorem backbone: a zero-free wall

Theorem 3.1 (SBM zero-free wall equivalence [SBM09]). Let $c, d > 0$ with $c \neq d$, and let b satisfy $1 > b \geq \frac{1}{2}$ and the admissibility hypotheses of SBM Theorem 1 (including the treatment of the limiting cases $b+c = 1$, $b+d = 1$). Then

$$\Delta_V(b; c, d) = 0 \iff \nexists \rho \text{ with } \operatorname{Re}(\rho) > b.$$

In particular,

$$\iff \Delta_V(\tfrac{1}{2}; c, d) = 0.$$

4 SFPT equilibrium as a global, coercive wall

Definition 4.1 (SFPT spectral fixed point). For fixed admissible (c, d) , define

$$\mathcal{B}_{c,d} := \{ b \in [\tfrac{1}{2}, 1) : \Delta_V(b; c, d) = 0 \}, \quad b_* := \inf(\mathcal{B}_{c,d} \cup \{1\}).$$

Proposition 4.2 (Fixed point equals the supremum of zero real parts). Let $\beta_0 = \sup_\rho \operatorname{Re}(\rho)$. For admissible (c, d) ,

$$b_* = \beta_0.$$

Proof. We consider two cases.

- **Case $\beta_0 = 1$:** For every $b < 1$, there exists a zero with $\operatorname{Re}(\rho) > b$ (by definition of β_0). Hence, by Theorem 3.1, $\Delta_V(b; c, d) \neq 0$. Therefore $\mathcal{B}_{c,d} = \emptyset$ and $b_* = \inf\{1\} = 1 = \beta_0$.
- **Case $\beta_0 < 1$:** If $b > \beta_0$, then no zero satisfies $\operatorname{Re}(\rho) > b$, so $\Delta_V(b; c, d) = 0$ by Theorem 3.1. If $b < \beta_0$, then there exists a zero with $\operatorname{Re}(\rho) > b$, so $\Delta_V(b; c, d) \neq 0$ by Theorem 3.1. Therefore $\mathcal{B}_{c,d} = [\beta_0, 1) \cap [\frac{1}{2}, 1) = [\beta_0, 1)$ and $b_* = \beta_0$.

□

Corollary 4.3 (RH as a global equilibrium).

$$\Longleftrightarrow b_* = \frac{1}{2}.$$

This is a statement of **global wall minimality** (“the first zero-free wall occurs at $1/2$ ”), not a symmetry-forced local stationarity condition.

5 Computational SFPT core: prime–zero truncation mismatch

The preceding sections are theorem-backed but “zero-side.” SFPT adds a computable proxy designed to converge to Δ_V .

5.1 Prime-side phase surrogate

Define a smoothed truncation using von Mangoldt weights:

$$\log \zeta_N(s) := \sum_{n \leq N} \frac{\Lambda(n)}{n^s \log n} w_N(n),$$

with a smoothing kernel w_N (e.g. Cesàro or Gaussian). Define $\arg \zeta_N(b + it)$ by continuous phase tracking along t .

5.2 Zero-side phase surrogate

Given zeros ρ with $|\operatorname{Im}(\rho)| \leq T$, define $\arg \zeta_T(b + it)$ via a truncated Hadamard/argument-principle representation (with the pole and Γ -factor handled consistently if one works with ξ instead of ζ).

5.3 SFPT computable free-energy divergence

Definition 5.1 (SFPT free-energy divergence).

$$\Delta_{N,T}(b; c, d) := \int_0^\infty W_{c,d}(t) \left| \arg \zeta_N(b + it) - \arg \zeta_T(b + it) \right|^2 dt.$$

5.4 SFPT convergence postulate (explicitly labeled)

Under appropriate smoothing, branch conventions, and coupled limits ($N, T \rightarrow \infty$),

$$\Delta_{N,T}(b; c, d) \longrightarrow \Delta_V(b; c, d).$$

This is the computational bridge: SFPT aims to approximate a known RH-equivalent obstruction by a measurable prime-zero mismatch, without assuming RH.

Proof sketch (contour $\rightarrow$ residues $\rightarrow$ zero-wall)

The identity (1) follows by applying the residue theorem to $I = \oint_C g(z) \log \zeta(z) dz$ with a rectangular contour C whose left edge is $\Re(z) = b$ and whose right edge is taken to the right of $b + c$ and $b + d$ (and then sent to ∞ in the SBM argument). On the left edge $z = b + it$ one has $g(b + it) = \frac{t}{(c^2 + t^2)(d^2 + t^2)} \in \mathbb{R}$, so taking the imaginary part of the contour identity produces the weighted integral of $\arg \zeta(b + it)$. The poles of $g(z)$ at $z = b + c$ and $z = b + d$ produce the first logarithmic term in $\mathcal{C}_{\text{expl}}(b; c, d)$, while the pole of ζ at $z = 1$ contributes the second logarithmic term (handled via the generalized Littlewood theorem as in [SBM09]). Each nontrivial zero ρ inside the contour contributes a term to $S(b; c, d)$, and in the limiting contour these are precisely the zeros with $\Re(\rho) > b$. Thus $\Delta_V(b; c, d) = |S(b; c, d)|$ measures the total obstruction coming from zeros lying to the right of the wall $\Re(s) = b$.

6 Numerical Methods

This section specifies a concrete, reproducible protocol for computing the theorem-backed divergence $\Delta_V(b; c, d)$ and the SFPT proxy $\Delta_{N,T}(b; c, d)$, together with grid rules, tolerances, and diagnostics designed to detect branch, truncation-tail, and insufficient-spectrum artifacts.

6.1 Canonical parameters and b -grid

Unless otherwise stated we fix

$$(c, d) = \left(\frac{3}{2}, \frac{7}{2}\right), \quad W_{c,d}(t) = \frac{t}{(c^2 + t^2)(d^2 + t^2)}.$$

We evaluate on a coarse grid $b \in \{0.55, 0.60, 0.65, 0.70, 0.75, 0.80\}$ and then refine locally near any observed transition or threshold in $\Delta_V(b; c, d)$ or

$\Delta_{N,T}(b; c, d)$. We avoid $b = \frac{1}{2}$ in initial numerics to prevent phase discontinuities at critical-line zeros.

6.2 Computing $\arg \zeta(b + it)$ without phase unwrapping

To avoid branch/unwrap discontinuities, we compute $\arg \zeta(b + it)$ from an ODE driven by the logarithmic derivative. For $s = b + it$,

$$\frac{d}{dt} \arg \zeta(b + it) = \Re \left(\frac{\zeta'}{\zeta}(b + it) \right). \quad (3)$$

We therefore define the continuous argument by

$$\arg \zeta(b + it) = (\zeta(b)) + \int_0^t \Re \left(\frac{\zeta'}{\zeta}(b + i\tau) \right) d\tau, \quad (4)$$

where $(\zeta(b))$ is the principal argument of the real number $\zeta(b)$ (i.e. 0 if $\zeta(b) > 0$ and π if $\zeta(b) < 0$); this pins the absolute phase at $t = 0$ and ensures continuity in t .

6.3 t -grid and quadrature

All t -integrals $\int_0^\infty (\dots) dt$ are truncated to $[0, T_{\max}]$ and evaluated with composite Simpson's rule on a piecewise-uniform grid:

$$\Delta t = \begin{cases} h_1 = 0.005, & t \in [0, 5], \\ h_2 = 0.02, & t \in (5, 50], \\ h_3 = 0.10, & t \in (50, T_{\max}]. \end{cases}$$

Default $T_{\max} = 300$. Tail stability is assessed by repeating the computation with $2T_{\max}$. (Heuristically $W_{c,d}(t) \sim t^{-3}$, so the weighted tail is typically mild, but we enforce this numerically.)

6.4 Gold-standard divergence $\Delta_V(b; c, d)$

For each b we compute the explicit constant

$$\mathcal{C}_{\text{expl}}(b; c, d) = \frac{\pi}{2(d^2 - c^2)} \left[\ln \left| \frac{\zeta(b + d)}{\zeta(b + c)} \right| + \ln \left| \frac{(d^2 - (1 - b)^2)c^2}{(c^2 - (1 - b)^2)d^2} \right| \right]. \quad (5)$$

We then compute $\arg \zeta(b + it)$ on the grid via (4) using high-precision evaluation of $\Re(\zeta'/\zeta)(b + it)$, and evaluate

$$I(b; c, d) := \int_0^{T_{\max}} W_{c,d}(t) \arg \zeta(b + it) dt. \quad (6)$$

Finally,

$$\Delta_V(b; c, d) := |\mathcal{C}_{\text{expl}}(b; c, d) - I(b; c, d)|. \quad (7)$$

Recommended tolerances. We accept a computed $\Delta_V(b; c, d)$ value only if all the following hold:

1. **Tail test:** $|I_{2T_{\max}} - I_{T_{\max}}| \leq \varepsilon_{\text{tail}}$ with $\varepsilon_{\text{tail}} = 10^{-6}$ (tighten to 10^{-8} for final plots).
2. **Mesh refinement:** halving each h_i changes $I(b; c, d)$ by at most $\varepsilon_{\text{mesh}} = 10^{-6}$.
3. **Precision test:** increasing working precision (e.g. $p \rightarrow p + 40$ digits) changes $I(b; c, d)$ by at most $\varepsilon_{\text{prec}} = 10^{-7}$.

6.5 SFPT proxy $\Delta_{N,T}(b; c, d)$

We define two continuous-phase surrogates $\arg \zeta_N$ (prime side) and $\arg \zeta_T$ (zero side), both normalized at $t = 0$ to $(\zeta(b))$.

Prime-side log-derivative surrogate. Fix a smoothing kernel; default is Gaussian

$$w_N(n) = \exp(-(n/N)^2).$$

Define

$$\left(\frac{\zeta'}{\zeta}\right)_N(s) := - \sum_{n \leq N} \Lambda(n) n^{-s} w_N(n). \quad (8)$$

Then set

$$\arg \zeta_N(b + it) := (\zeta(b)) + \int_0^t \Re \left(\frac{\zeta'}{\zeta}\right)_N(b + i\tau) d\tau. \quad (9)$$

Zero-side log-derivative surrogate. Let $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$. Using a zero list $\{\rho : |\Im(\rho)| \leq T\}$, approximate

$$\left(\frac{\xi'}{\xi}\right)_T(s) \approx \sum_{|\Im(\rho)| \leq T} \frac{1}{s - \rho}. \quad (10)$$

Then define

$$\left(\frac{\zeta'}{\zeta}\right)_T(s) := \left(\frac{\xi'}{\xi}\right)_T(s) - \frac{1}{s} - \frac{1}{s-1} + \frac{1}{2} \log \pi - \frac{1}{2} \frac{\Gamma'}{\Gamma}\left(\frac{s}{2}\right), \quad (11)$$

and set

$$\arg \zeta_T(b + it) := (\zeta(b)) + \int_0^t \Re\left(\frac{\zeta'}{\zeta}\right)_T(b + i\tau) d\tau. \quad (12)$$

SFPT proxy divergence. With these surrogates we compute

$$\Delta_{N,T}(b; c, d) := \int_0^{T_{\max}} W_{c,d}(t) \left| \arg \zeta_N(b + it) - \arg \zeta_T(b + it) \right|^2 dt. \quad (13)$$

Recommended schedules and tolerances. We recommend

$$N \in \{10^4, 10^5, 10^6\}, \quad T \in \{T_{\max} + 50, 2T_{\max} + 50\},$$

with acceptance conditioned on:

1. **Normalization check:** $\arg \zeta_N(b) = \arg \zeta_T(b) = (\zeta(b))$ (to numerical roundoff).
2. **Monotone improvement:** for fixed b , $\Delta_{N,T}(b; c, d)$ should decrease (or at least stabilize) as N and T increase past a transient regime.
3. **Tail test:** doubling $T_{\max}$ changes $\Delta_{N,T}$ by at most $\varepsilon_{\text{tail}}^{(2)} = 10^{-6}$.

6.6 Diagnostic checklist

Table 1 lists common failure modes and how to resolve them.

References

- [SBM09] S. K. Sekatskii, S. Beltraminelli, D. Merlini. *A few equalities involving integrals of the logarithm of the Riemann zeta-function and equivalent to the Riemann hypothesis II.* arXiv:0904.1277 (2009).
- [Vol95] V. V. Volchkov. *On an equality equivalent to the Riemann hypothesis.* Ukrainian Mathematical Journal **47** (1995), 491–493.

Symptom	Likely cause	Fix / test
$\arg \zeta(b+it)$ exhibits jumps of size $\approx 2\pi$	Branch/unwrap bug (phase tracking)	Use (4) (integrate $\Re(\zeta'/\zeta)$) instead of direct arg evaluation; verify continuity on a dense grid.
Δ_V or $I(b; c, d)$ changes materially when $T_{\max}$ doubles	Tail truncation bug	Increase $T_{\max}$; enforce $ I_{2T_{\max}} - I_{T_{\max}} \leq \varepsilon_{\text{tail}}$; consider extending the coarse-grid region with larger $T_{\max}$.
$\Delta_{N,T}$ does not improve as T increases	Insufficient T (too few zeros / spectral truncation)	Increase T (at least $T \geq 2T_{\max}$); spot-check $\arg \zeta_T$ against a high-precision direct $\arg \zeta$ at random t -points.
$\Delta_{N,T}$ does not improve as N increases	Insufficient N or poor smoothing	Increase N ; use smoother w_N (Gaussian default); verify stability under modest changes of the smoothing scale.
Results change when increasing working precision	Precision bug / cancellation	Raise precision (e.g. $p \rightarrow p + 40$ digits) until $\varepsilon_{\text{prec}}$ is met; avoid subtracting nearly equal large quantities without extra precision.
Mesh halving changes outputs significantly	Quadrature / grid resolution bug	Halve Δt in each region; verify Simpson stability; densify near t where $\Re(\zeta'/\zeta)$ is large.

Table 1: Numerical diagnostics for the SBM-based SFPT 2.0 implementation.